

and structure of terrain required, a greyscale height-map is generated, known as the displacement map. The Parhelia uses this displacement map, along with a low-polygonal base-mesh to generate a completely 3D terrain environment. The same procedure can also be used to generate 3D models. Since HDM involves relatively small files to generate a complex, high-polygon-count model, the technique is a great bandwidth saver and a very efficient solution. Matrox is working hard to get this feature included in both DirectX 9 and OpenGL 2.0.

■ **Surround Gaming:** Picture a desktop with three monitors. You fire up *Jedi Knight II* to sabre-fry a Bantha. The monitor in the

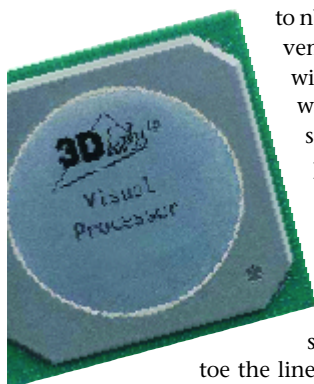


Parhelia's surround gaming bridges the gap between virtual-reality and reality

centre gives you the normal view, while the two adjacent ones, show you what to expect on your left and right hand sides. All three combine to give you a complete picture along with a peripheral view. The technology will serve you even better in a flight-simulation game, where positional awareness can mean the difference between life and death. This is Surround Gaming, and the technology can well extend to your office work, where you can browse the Internet, work in Photoshop and write a Word document, all on separate monitors running on the same PC.

3Dlabs' P10

Creative has long presented consumers with 3D solutions based on chips ranging from 3dfx's Voodoo Banshee to nVidia's GeForce. Its ventures have met with limited success, while its relationships with third-party video chip providers have always been shaky at best. Nevertheless, Creative seemed content to toe the line. Until it acquired



3Dlabs earlier this year, that is.

3Dlabs was working on a new 3D chip, and the powers-that-be at Creative were so impressed by the technology behind the chip and the promise it held in terms of market penetration, that they decided to buy 3Dlabs outright. This decision gives Creative a degree of control over when and how the chip is developed. The relationship will work along these lines: 3Dlabs will continue as designers and developers, while Creative will market video cards based on these very chips, all the while lending a helping hand in tasks ranging from driver development to marketing.

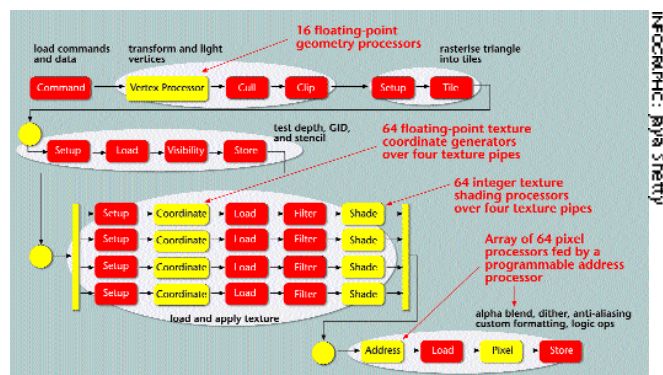
When a CPU meets a GPU

Developments in the CPU industry and the graphics chips industry have followed some amazing parallels. In the early days of Voodoo cards, every 3D feature, every neat trick was hardwired inside the chips. This fixed-functionality brought amazing speeds but severely hampered innovation and creativity. Then nVidia with its GeForce 256 chip and later on ATi with the Radeon 7500, introduced bare-bone programmability in their respective offerings. It was like moving from a calculator that could only perform certain tasks, to the early Intel chips that allowed for assembly programming. The GeForce 3 and 4, and the Radeon 8500 have taken the next logical step, extending the programmability from the vertex level to the pixel level—but the gaming industry is demanding more: a completely programmable graphics processor, to be precise.

This is where 3Dlabs has stepped in. Although the P10, the latest chip to come from the 3Dlabs drawing board, is not a completely programmable solution, it is as close to the real deal as current manufacturing technologies can allow for.

Innovations throughout

So just how programmable is this new chip? There are over 200 32-bit floating-point and integer processors built into the P10. These are all general purpose processors, with



The P10 has 16 floating point geometry processors that work in parallel. These feed data through the transform and lighting data path. The texture processor contains 128 32-bit SIMD processors (64+64), enabling the P10 to process eight simultaneous textures. Yet another set of processors is available for additional pixel operations. All of this makes the P10 capable of multithreading

addressing, looping and subroutine capability. All programmable, together they have earned the chip a new acronym: VPU, short for Visual Processor Unit.

The P10 has 16, separate 32-bit floating-point geometry processors, which handle the vertex processing. This array of 16 vertex processors is the P10's equivalent of vertex shaders and they offer full DirectX 8 and 9 support. Also noteworthy is that the final stage of the rendering pipeline is also completely programmable. What this means is that developers can apply custom filtering methods if the standard bilinear/trilinear filtering doesn't suit their needs. Also programmable is the type and the extent of antialiasing that can be applied to the final output. Because of this final rendering stage, the P10 can be programmed to support greater colour depths, offering even 64-bit colour. The VPU from 3Dlabs also has 10-bit DACs to take advantage of the custom 10-10-10-2 (RGBA) mode.

Then there's the multi-threaded Command Processor that enables the P10



The accuracy provided by a 64-bit colour depth has helped movie companies such as Pixar bring their digital worlds to life. Programmable chips like the P10 will allow game developers to have a similar level of visual quality at minimal performance hit